

18. Answer: C

$$\text{Using } E = \frac{Q}{4\pi\epsilon_0 r^2}, E \propto \frac{1}{r^2},$$

$$\frac{E_{-q}}{E_{+q}} = \left(\frac{0.30}{0.60} \right)^2 = \frac{1}{4}$$

$$E_{-q} = \frac{1}{4} E_{+q}$$

$$= \frac{1}{4} \times 10$$

$$= 2.5 \text{ N C}^{-1} \text{ (towards charge } -q)$$

$$\text{Magnitude of the electric field strength} = 10.0 + 2.5 = 12.5 \text{ N C}^{-1}$$